NewsBot Intelligence System: Reflective Journal

Student: Dana Bolden
Course: ITAI 2373 – Mid-Term Project

Team Collaboration Analysis

I completed this project on my own, so the usual team strategies didn't apply. Instead of dividing the eight NLP modules among teammates, I managed the work like a small software project, with a clear order of tasks and a time limit for each stage. Dataset preparation and text cleaning (Modules 1 and 2) came first, since everything else depended on them. Once the text was normalized and the data structures were in place, I worked in short sprints: the analytical modules (TF-IDF, part-of-speech tagging, and syntax) first, followed by the machine learning modules (sentiment analysis, classification, and named entity recognition). Working alone meant I had to be disciplined about version control and test each stage before moving on, so that every downstream model received clean, correctly formatted input.

Technical Integration Challenges

Getting spaCy, NLTK, and scikit-learn to work together in a single pipeline was harder than I expected. Most of the problems came down to data formatting, environment stability, and feature scaling.

The first issues appeared during setup in Google Colab, where missing NLTK resources (`punkt_tab` and `averaged_perceptron_tagger_eng`) caused deprecation and environment errors. I fixed these by adding manual downloads to the setup script.

The bigger challenge came in Module 7 (Classification). When I combined the TF-IDF matrix with sentiment and text-length features, the Naive Bayes classifier crashed because VADER produces negative sentiment scores, which that algorithm cannot accept. I rescaled the sentiment features from a range of [-1, 1] to [0, 1], which resolved the error. A second problem followed: the raw character counts were so much larger than the TF-IDF values that they overwhelmed the word vectors, and accuracy stalled at around 71%. Applying a MinMaxScaler to the length features put everything on a comparable scale, and Naive Bayes accuracy jumped to 97.32%.

Business Value Assessment

The NewsBot Intelligence System shows clear commercial potential for media monitoring, PR tracking, and news aggregation. After processing 1,490 articles, it handled the triage of unstructured text reliably. At 97.32% accuracy, a publisher could reasonably use the classifier to route incoming articles to the right editorial desks, such as Tech, Sport, or Business, without sorting them by hand.

The system also extracted 44,930 named entities, which gave each category a distinct profile. Business and Politics coverage leaned heavily on organizations (ORG) and financial figures (MONEY), while Sports and Entertainment were driven mainly by people (PERSON). The sentiment module added another layer: Entertainment was the most positive category (+0.5865), while Politics scored lowest (+0.0360), which is close to neutral. Together, these modules would let a business track what is being said about a company and how audiences are reacting to it.

Individual Contributions

As the only developer, I handled the project from start to finish, including the architecture, implementation, and analysis. Specifically, I:

- Acquired and cleaned the Kaggle dataset
- Built the text preprocessing pipeline to remove noise
- Generated the TF-IDF vector matrices
- Ran grammatical and syntactic parsing
- Trained, evaluated, and tuned the machine learning classifiers, optimizing the Naive Bayes model for the best accuracy
- Wrote all business insights, translating the statistical results into practical recommendations

Future Enhancements

The current NewsBot relies on traditional machine learning and statistical NLP. The most valuable next step would be moving the classification and NER modules to Transformer-based models such as BERT or RoBERTa. These models capture context and nuance, including sarcasm, that TF-IDF representations miss.

I would also add a temporal analysis component. By connecting the pipeline to a live news API, the system could track how sentiment toward a specific entity, such as a political figure or a tech company, changes over time.

Professional Development Impact

Building NewsBot from end to end changed how I think about my work. I used to write isolated code snippets; now I think in terms of complete data pipelines. Turning messy Kaggle data into structured business insight forced me to think carefully about system design and resource management.

The most useful lesson was learning to present technical results, such as an F1-score or a POS distribution, in terms of what they mean for a business decision. This project is a practical portfolio piece that shows I can apply machine learning to real-world problems.